BENCHAPHORN CHANPRASERTKUL

Personal Information

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Education

King Mongkut's University of Technology Thonburi
Bachelor of Engineering in Mechatronics Engineering, Second-Class Honors | 2020–2024

- **Junior Science Talent Project (JSTP-SCB) Scholar** by the National Science and Technology Development Agency(NSTDA)

Professional Summary

Software Engineer with experience in Python, Linux, Docker, cloud infrastructure, CI/CD, and web/mobile application development. Former CERN Summer Student with hands-on experience improving large-scale testing infrastructure using GitHub Actions, Docker, and OpenStack. Experienced in software design, testing, AWS-based systems, and collaborating with international customers and engineering teams. Interested in distributed computing, workflow management, and large-scale scientific software systems.

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL, YAML, HTML/CSS

Systems & Infrastructure: Linux, Docker, OpenStack, AWS

CI/CD & Development: Git, GitHub Actions, Bitbucket Pipelines, GitHub Artifacts, SonarCloud/SonarQube

Databases: MySQL, DynamoDB, MongoDB

Frameworks: Angular, React Native, Node.js, NestJS, Flask

Testing: pytest, Maestro, unit testing

Work Experience

Toyota Tsusho NEXTY Electronics (Thailand) Co., Ltd. **Bangkok, Thailand**
Software Engineer Feb 2025 - Present

- Design, develop, test, and maintain software for IoT/smart-home systems integrated with AWS cloud services.
- Develop and maintain **Python-based tools and automated tests** supporting software development and system validation.
- Analyze software requirements and translate customer requirements into technical designs and implementation specifications.
- Investigate application, API, cloud, and device integration issues across distributed system components.
- Improved **CI/CD pipelines** by optimizing test execution and reducing unnecessary processing time.
- Conduct software security testing and improve application security configurations.
- Review software requirements, implementations, and test specifications to maintain software quality.
- Communicate technical requirements, implementation plans, issues, and results with international customers in **English.**
  - **Technologies:** Python, JavaScript, React Native, Angular, AWS, DynamoDB, Git, Bitbucket Pipelines, SonarCloud
  - **Tools:** Bitbucket, Maestro, Jira, Confluence

Internship Experience

European Organization for Nuclear Research, CERN — **Geneva, Switzerland**

DevOps - CERN Summer Student 2024 — June - Aug 2024

Responsible: Testing Infrastructure with GitHub Action.

- Developed and optimized Python-based testing infrastructure for the Xsuite project using GitHub Actions.
- Improved CI scalability by parallelizing test execution across multiple runners.
- Designed a workflow to build and reuse pre-built Docker images containing dependencies, GPU drivers, and Xsuite packages, reducing redundant builds and improving CI efficiency.
- Worked in a Linux and OpenStack environment with Docker, GitHub Actions, GitHub Artifacts, and Git
  - **Technologies:** Python, Linux, Docker, GitHub Actions, Git, OpenStack
  - **GitHub:** https://github.com/SuuGarr/xsuite

MyOrder company — **Remote, Thailand**

Full-stack developer — June - July 2023

Responsible: Developed web applications.

- Developed web applications for a backend e-commerce management system, implementing features to manage online store operations and monitor seller order volumes.
  - **Technologies:** JavaScript, TypeScript, CSS, HTML, Angular, NestJS, Node.js, MongoDB (NoSQL database), Docker, SonarQube, Git

The National Electronics and Computer Technology Center (NECTEC) — **Bangkok, Thailand**

Full stack developer — June - July 2022

Responsible: Developed web applications.

- Developed a LINE chatbot with GPS, camera, and database integration for facial authentication.
- Developed a web application for AI-based face spoof detection using human behavior analysis.
  - **Technologies**: JavaScript, CSS, HTML, Python, MySQL

Certificate and Awards

Predictive Control of Aluminum Welding Using Infrared Thermal Imaging and Machine Learning

- **Best Dissertation Project –** Chalit Industry Award 2024
- **Best Oral Presentation –** ISPC Conference 2024
- Developed a machine-learning system for welding crack detection and optimal parameter prediction using Python, infrared thermal imaging, image processing, and YOLOv8.

Machine Condition Monitoring – Delta Cup 2023

- Second Runner-up – Oral Presentation
- Developed a machinery condition-monitoring system using vibration and temperature sensors, FFT signal analysis, cloud processing, and predictive analysis.

Intelligent Prosthetic Arm

- Silver Medal – Thailand Research Expo 2022
- Honorable Mention – 24th National Software Contest (NSC 2022)
- Developed a sensor-controlled prosthetic arm using flex sensors to detect muscle-area bending signals and translate them into prosthetic arm movement.

Certifications & Training

- **Google Cloud Fundamentals:** Core Infrastructure – Trainocate, 2022
- **Web Application Penetration Testing –** Mahidol University & depa Thailand, 2023